LC-MS Method Report

Test Method with Long SMILES — Generated on September 05, 2026 at 03:35 UTC

Compound Information

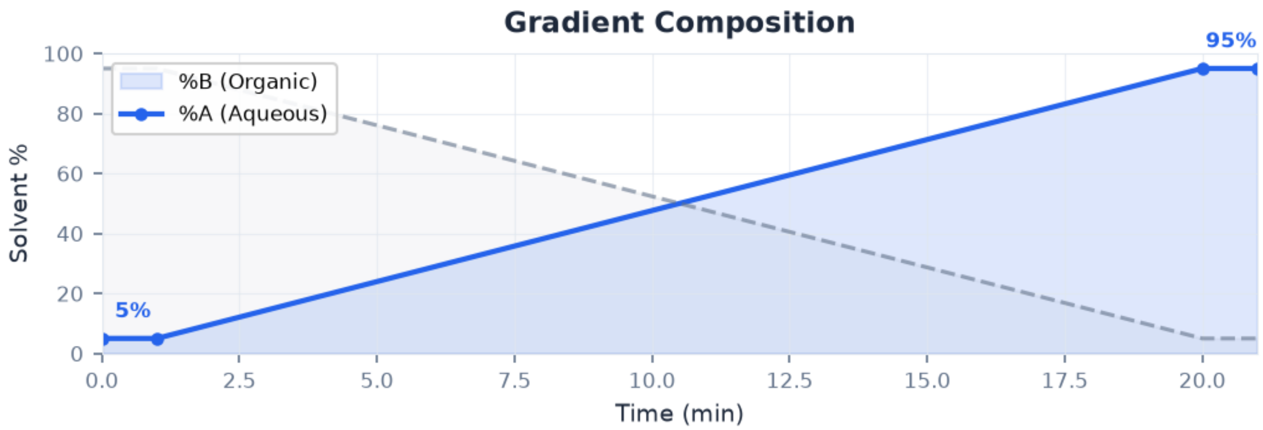
Name	Caffeine
SMILES	`CN1C=NC2=C1C(=O)N(C(=O)N2C)C`
InChIKey	`RYYVLZVUVIJVGH-UHFFFAOYSA-N`

Property	Value	Property	Value
MW (g/mol)	194.19	HBD	—
logP	-0.07	HBA	6
TPSA (Å²)	58.4	Rotatable Bonds	—
pKa (est.)	14.0	Aromatic Rings	2

Method Parameters

Parameter	Value	Parameter	Value
Column Type	C18	pH	2.70
Mobile Phase A	Water + 0.1% Formic Acid	Flow (mL/min)	0.40
Mobile Phase B	Acetonitrile	Temp (°C)	30
Additive	0.1% Formic Acid	Column Length	100 mm

Gradient Program

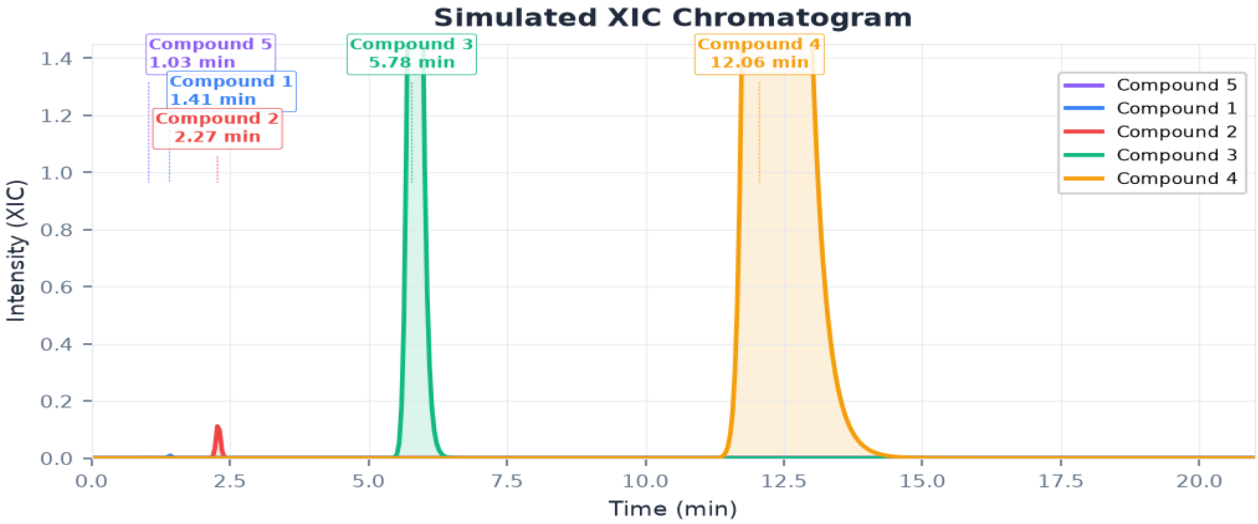


Time (min)	%B	%A
0.00	5.0%	95.0%
1.00	5.0%	95.0%
20.00	95.0%	5.0%

Predictions are estimates derived from physicochemical heuristics and statistical models. They require experimental verification before use in regulated or production analytical work.

21.00	95.0%	5.0%

Simulated XIC Chromatogram



#	Compound	RT (min)	Width (s)	Tailing
1	Compound 1	1.41	3.1	1.34
2	Compound 2	2.27	5.0	1.43
3	Compound 3	5.78	12.7	1.78
4	Compound 4	12.06	26.4	2.41
5	Compound 5	1.03	2.3	1.30

Resolution Matrix (Rs)

Compound	Compound 5	Compound 1	Compound 2	Compound 3	Compound 4
Compound 5	—	8.41	20.56	38.17	46.13
Compound 1		—	12.89	33.32	43.32
Compound 2			—	23.85	37.39
Compound 3				—	19.26
Compound 4					—

Rs > 1.5 indicates baseline resolution. Rs < 1.5 (marked with !) indicates co-elution risk.

Disclaimer

Predictions are estimates derived from physicochemical heuristics and statistical models. They require experimental verification before use in regulated or production analytical work. pKa values are heuristic estimates. Retention times may differ from actual experimental results due to column batch variability, instrument configuration, and sample matrix effects.